

Essential Topics for Selenium

1. Python Basics

- Variables and data types (str, int, float, bool)
- Input/output (input(), print())
- Operators (arithmetic, logical, comparison)

2. Control Structures

- if, elif, else
- for and while loops
- break and continue

3. Functions

- Defining and calling functions (def)
- Arguments and return values
- lambda functions (optional but useful)

4. Data Structures

- Lists (indexing, slicing, list methods)
- Dictionaries (keys, values, items)
- Tuples and Sets (basic usage)

5. String Manipulation

- Common string methods (`split()`, `strip()`, `replace()`, etc.)
- f-strings and string formatting

6. File Handling

- Reading from and writing to text/CSV files
- Basic understanding of file paths

7. Exception Handling

- `try`, `except`, `finally`
- Handling specific exceptions (like `FileNotFoundError`, `ValueError`)

8. Modules and Packages

- Importing built-in modules (`os`, `time`)
- Installing and using external packages (e.g., `pip install selenium`)

9. Object-Oriented Programming (OOP) – *Optional but Recommended*

- Classes and objects
- Attributes and methods
- Inheritance (if your Selenium use involves page object model)

10. Basic Understanding of HTML and XPath/CSS Selectors

- Tags, attributes, and DOM structure
- Using developer tools (Inspect element)
- Writing simple XPath and CSS selectors

Then Start with Selenium Topics

Once they know the basics of Python, you can begin teaching:

- Setting up Selenium and browser drivers
- Finding elements (by id, name, class, xpath, css_selector)
- Performing actions (click, send_keys, clear, etc.)
- Waits (time.sleep, WebDriverWait)
- Handling alerts, iframes, and multiple tabs
- Taking screenshots and scraping data
- Pytest
- Reporting
- POM